# FFSD PROJECT

## (1st Evaluation Documentation)

**Domain : Smart City & Urban Services Platforms**

**Problem Statement : Urban Service Request & Issue Management Platform**

**Team Name : Urbanity**

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# Software Requirements Specification (SRS)

## Urbainity – Urban Service Request & Issue Management Platform

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# 1. Preface

This document is intended for project evaluators, faculty members, developers, and other stakeholders involved in the design, development, and assessment of the Urban Service Request and Issue Management Platform. It provides a structured and comprehensive description of the system requirements to ensure a shared understanding among all readers.

This Software Requirements Specification (SRS) document corresponds to version 1.0 of the project. The primary objective of this version is to capture the complete set of functional and non-functional requirements of the system at an early stage of development. It serves as a baseline for system design, implementation, testing, and future enhancements.

This version has been created to formally document system expectations, user interactions, and operational constraints identified during the initial requirements analysis phase. Future versions of this document may incorporate refinements based on stakeholder feedback, changes in project scope, or evolving system needs.

# 2. Introduction

## 2.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define the functional and non-functional requirements of the Urban Service Request and Issue Management Platform. This document serves as a common reference for developers, project mentors, evaluators, and stakeholders to understand the system's objectives, expected behavior, and constraints.

## 2.2 Scope of the Project

The Urban Service Request and Issue Management Platform is a web-based application developed to address the need for efficient handling of civic issues in urban environments. The system enables citizens to report public service issues, track their status, and receive updates, while allowing municipal authorities to manage, assign, and resolve complaints in a structured manner.

The platform supports Smart City objectives by promoting transparent governance, improved service delivery, and active citizen participation through digital means. It provides centralized monitoring, accountability, and data-driven insights for urban service management.

The system may interact with external services such as mapping systems for location-based complaint reporting, notification services for status updates, and dashboard components for analytics and performance monitoring. It is designed to integrate smoothly with existing urban administration workflows while remaining scalable for future enhancements.

## 3. Glossary

| Term | Description |
|---|---|
| Citizen | A registered user who submits complaints or service requests |
| Complaint | A reported civic issue related to public infrastructure or services |
| Service Request | A formal request raised by a citizen for an urban service |
| Issue Category | Classification of a complaint based on the type of problem |
| Priority Level | Indicates the urgency of a complaint (Low, Medium, High) |
| Status | The current state of a complaint (Submitted, Assigned, In Progress, Resolved, Closed) |
| Department Officer | Official responsible for reviewing and managing complaints |
| Field Worker | Personnel assigned to inspect or resolve complaints on the ground |
| Department Head | Authority overseeing departmental operations and performance |
| Admin | System administrator managing users, roles, and system settings |
| Assignment | The process of allocating a complaint to an officer or field worker |
| Escalation | Forwarding an unresolved complaint to a higher authority |
| Resolution | Action taken to successfully address and close a complaint |
| Feedback | Response provided by a citizen after issue resolution |

| | |
|---|---|
| Notification | System-generated alert sent to users regarding updates |
| Dashboard | Interface displaying complaint statistics and system status |
| Location Data | Geographic information associated with a complaint |
| Resolution Time | Duration from complaint submission to closure |

# 4. Description

## 4.1 Product Perspective

The system is a standalone web application developed using a full-stack architecture. It consists of a frontend interface for users, a backend server for business logic and processing, and a database for persistent data storage. The platform may integrate with external services such as map APIs, notification services, or analytics tools.

## 4.2 User Classes and Characteristics

| User Type | Description |
|---|---|
| Citizen | Submits complaints and tracks their status |
| Department Officer | Reviews and assigns complaints |
| Field Worker | Performs on-ground resolution tasks |
| Department Head | Monitors department performance |
| Admin | Manages system settings and user roles |

## 4.3 Operating Environment

The platform is accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Brave, Safari,  and Microsoft Edge. No special software installation is required. The application runs on a centralized server or cloud infrastructure, with all data securely stored in a database system.

## 4.4 Design and Implementation Constraints

- Internet connectivity is required to access the system
- The platform must comply with basic data security practices
- System performance depends on server and network availability

# 5. User Requirements

## 5.1 Citizen Requirements

- Citizens shall be able to register and log in securely
- Citizens shall be able to submit complaints with descriptions and location details
- Citizens shall be able to view the status of their submitted complaints
- Citizens shall receive notifications on status updates
- Citizens shall be able to provide feedback after issue resolution

## 5.2 Department Officer Requirements

- Officers shall be able to log in securely
- Officers shall view and manage complaints assigned to their department
- Officers shall assign complaints to field workers
- Officers shall update the  complaint status

## 5.3 Field Worker Requirements

- Field workers shall view the assigned tasks
- Field workers shall update progress and resolution details
- Field workers shall upload proof of work if required

## 5.4 Department Head Requirements

- Department heads shall monitor complaint statistics
- Department heads shall review departmental performance
- Department heads shall escalate unresolved issues

## 5.5 Admin Requirements

- Admin shall manage users and roles
- Admin shall configure system settings
- Admin shall monitor overall system activity

# 6. System Architecture

The Urban Service Request and Issue Management Platform follows a modular, web-based architecture designed to support scalability, maintainability, and clear separation of responsibilities. The system is organized into distinct components, each responsible for a specific set of functions.

The **frontend module** provides the user interface for all system users, including citizens, department officers, field workers, department heads, and administrators. It enables users to interact with the system for activities such as complaint submission, status tracking, task assignment, and monitoring through dashboards. The frontend is accessible through standard web browsers and focuses on usability and responsiveness.

The **backend services** handle the core application logic and workflow management. These services process user requests, manage authentication and authorization, control the complaint lifecycle, and enforce business rules such as assignment and escalation. Backend services act as the central coordination layer between the user interface, the database, and external services.

The **database component** is responsible for persistent storage of system data, including user profiles, complaints, assignments, status updates, and feedback. It ensures data consistency, integrity, and reliable retrieval of information required by the application.

The system supports **external integrations** to enhance functionality and user experience. Mapping services may be used to capture and visualize location-based complaint data. Notification services may be used to deliver updates to users regarding complaint status changes. Dashboard and analytics components may leverage reusable visualization libraries or services.

Where appropriate, existing and proven services such as authentication mechanisms and map APIs may be reused to reduce development effort and improve reliability. The architecture is designed to allow future integration with additional services without major structural changes.

# 7. System Requirements

- Role-based access control(RBAC) must be supported for all users.
- User inputs must be validated before processing or storage.
- Duplicate complaints for the same issue and location should be prevented.
- Important system actions must be recorded in audit logs.
- Authorized users should be able to update the complaint priority and category.
- Complaint escalation should be supported based on predefined rules.
- Notifications must be generated for important status updates.
- Timestamps must be maintained for complaint creation, assignment, and resolution.
- Administrative users should be able to enable or disable platform features.
- Search and filter options must be available for managing complaints.
- Complaint submissions should support attachments such as images or documents.
- Data consistency must be maintained across all modules.
- Recovery mechanisms should be available to handle system failures.
- Basic analytics should be provided to monitor complaint trends.
- The architecture should allow future integration with external services.

# 8. Functional Requirements

## 8.1 Authentication

- Users must be able to register and log in securely.
- User credentials must be validated during authentication.
- Access to features must be controlled based on assigned user roles.

## 8.2 Complaint Management

- Citizens must be able to submit complaints through the platform.
- Submitted complaints must be routed to the appropriate departments.
- The status of complaints must be updated as actions are performed.

## 8.3 Monitoring and Reporting

- Dashboards must be available for department officers and administrators.
- Reports on complaint resolution and system activity must be generated.

# 9. Non-Functional Requirements

## 9.1 Performance

- The system shall respond within acceptable time limits
- The system shall support multiple concurrent users

## 9.2 Security

- User passwords shall be encrypted
- Secure communication using HTTPS
- Unauthorized access shall be prevented

## 9.3 Usability

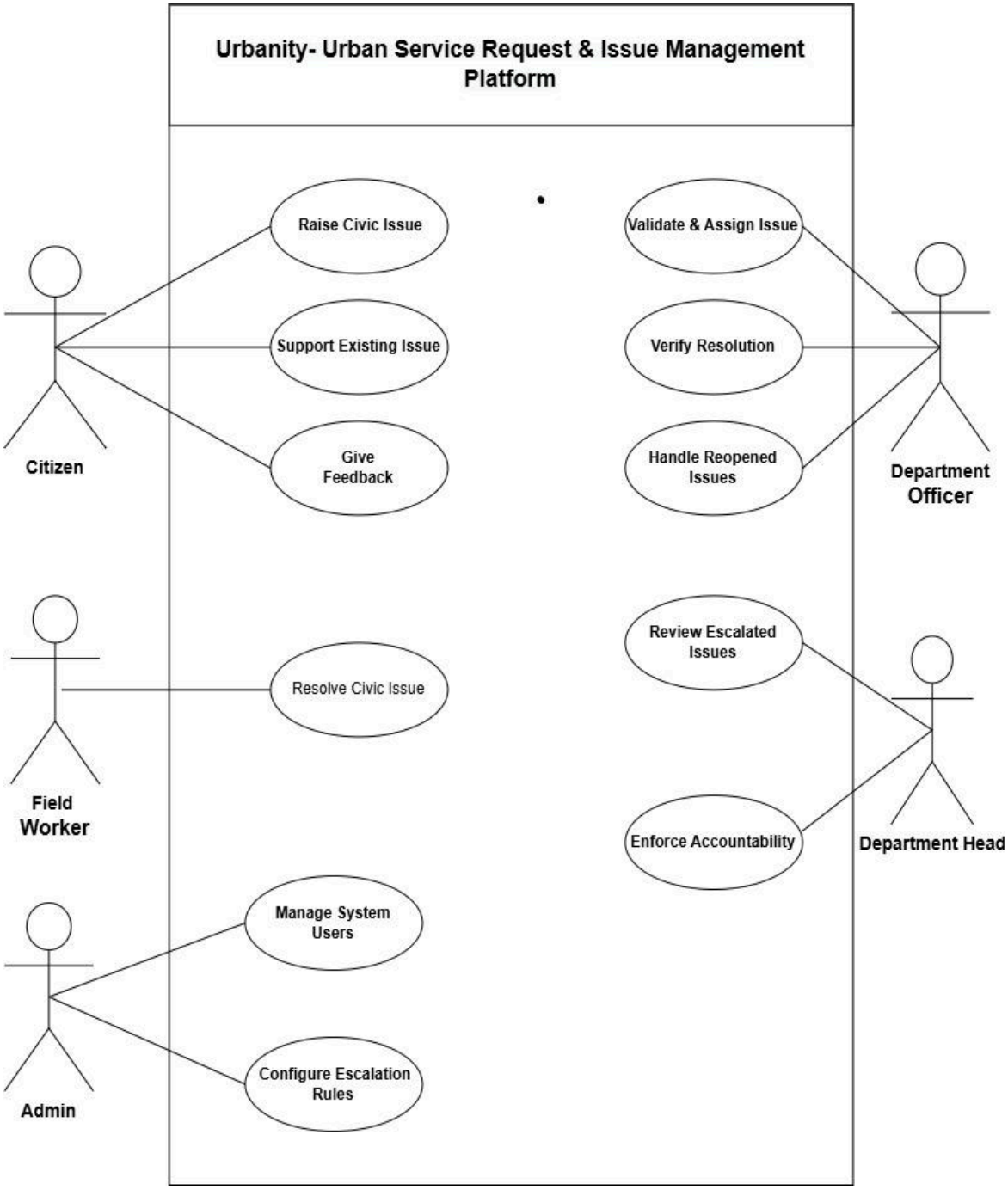
- The system shall be easy to use for non-technical users
- The interface shall be responsive for desktop and mobile devices

## 9.4 Reliability

- The system shall ensure high availability
- The system shall handle failures gracefully

## 9.5 Scalability

- The system shall support future feature enhancements
- The system shall handle increased user load

# 10. System Models

10.1 Use Case Diagram: